

Model Fairness Audit: German Credit - Logistic Regression

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1. Executive summary

100_{/100}
Fairness score

Verdict: **Pass**

Primary metric: Equal opportunity. Are truly qualified people approved at similar rates?

Pass threshold: True-positive-rate ratio of at least 0.80.

Key findings

- No group with adequate sample size fails equal opportunity.
- For context, the lowest selection-rate ratio is 0.724 (the four-fifths rule requires 0.80).
- Overall drift is stable (highest PSI 0.0773); 0 feature(s) shifted.
- Decision explanations reproduce the model's decisions on 100% of records.
- 'age' is strongly associated with 'age_band' (Cramer's V 0.98) and may act as a proxy.

Priority actions

- **Medium** 'age' carries essentially the same information as 'age_band' (Cramer's V 0.98). If the model uses it, the sensitive attribute is effectively a direct input; confirm this is lawful and justified for the use case.
- **Low** 7 group(s) have fewer than 100 people; their results are low-confidence.
- **Low** Schedule a fairness and drift re-check every quarter.

2. Scope: data and model

Model	German Credit - Logistic Regression
Intended use	Lending
Records audited	400
Columns	23
Model decision column	predicted_good_credit
Actual outcome column	good_credit
Sensitive attributes	sex, age_band
Overall selection rate	76%
Overall accuracy	76%
Positive outcome base rate	70%

3. Method and metric selection

Records are grouped by each sensitive attribute and by every combination of two or more attributes (intersectional groups). Each group is compared with a reference group: the best-off group with at least 100 records. The primary metric, **Equal opportunity**, decides pass or fail; the other metrics are reported for context.

Metric	Question answered	Pass threshold
Demographic parity	Are groups selected at similar rates?	Selection-rate ratio of at least 0.80
Equal opportunity	Are truly qualified people approved at similar rates?	TPR ratio of at least 0.80
Equalized odds	Are misses and false alarms similar across groups?	TPR and FPR gaps of at most 0.10
Predictive parity	Does a positive decision mean the same for every group?	Precision ratio of at least 0.80

Groups below 100 records are marked low confidence; if they fail they are marked "Review" rather than "Fail". Selection rates include 95% Wilson confidence intervals. The fairness score blends average and worst-group compliance with the primary metric, minus a penalty per failing group, and is capped below 70 whenever any group fails.

4. Fairness results: single attributes

sex

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
male (reference)	283	77% (72%-82%)	1.00	87%	53%	1.00	0.00	1.00	1.00	Pass
female	117	71% (62%-78%)	0.92	87%	43%	1.00	0.10	0.96	1.00	Pass

age_band

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
45plus	83	78% (68%-86%)	1.00	95%	38%	1.10	0.17	1.05	1.10	Pass
25-44 (reference)	237	78% (72%-83%)	1.00	86%	55%	1.00	0.00	1.00	1.00	Pass
under25	80	65% (54%-75%)	0.83	80%	47%	0.93	0.08	0.82	0.93	Pass

5. Fairness results: intersectional groups

sex x age_band

Group	Size	Selection rate (95% CI)	Selection ratio	TPR	FPR	TPR ratio	Odds gap	Precision ratio	Equal opportunity	Status
female, 45plus	21	86% (65%-95%)	1.10	94%	50%	1.10	0.09	1.07	1.10	Pass
male, 25-44 (reference)	187	78% (72%-83%)	1.00	85%	56%	1.00	0.00	1.00	1.00	Pass
female, 25-44	50	78% (65%-87%)	1.00	89%	53%	1.04	0.03	0.96	1.04	Pass
male, under25	34	76% (60%-88%)	0.98	81%	69%	0.95	0.14	0.79	0.95	Pass
male, 45plus	62	76% (64%-85%)	0.97	95%	35%	1.12	0.21	1.03	1.12	Pass
female, under25	46	57% (42%-70%)	0.72	78%	35%	0.92	0.21	0.83	0.92	Pass

6. Drift analysis

Population Stability Index (PSI): below 0.10 stable, 0.10 to 0.25 moderate, above 0.25 significant. The Kolmogorov-Smirnov p-value is given for numeric features (below 0.05 indicates a significant shift).

Feature	PSI	KS p-value	Band
credit_amount	0.077	0.1643	Stable
savings	0.036	n/a	Stable
purpose	0.034	n/a	Stable
credit_history	0.026	n/a	Stable
age	0.024	0.6388	Stable
employment	0.018	n/a	Stable
property	0.016	n/a	Stable
other_debtors	0.015	n/a	Stable
existing_credits	0.014	0.3953	Stable
duration	0.014	0.9945	Stable
residence_since	0.014	0.8629	Stable
job	0.013	n/a	Stable
housing	0.007	n/a	Stable
status	0.007	n/a	Stable
telephone	0.002	n/a	Stable
installment_rate	0.002	1.0	Stable
other_installment	0.000	n/a	Stable
foreign_worker	0.000	n/a	Stable
num_dependents	0.000	1.0	Stable

7. Explainability

Explanations are produced by a transparent surrogate model that reproduces the audited model's decisions on **100%** of records. Sensitive attributes are excluded from explanations and counterfactuals.

Overall drivers of the model's decisions

Feature	Share of influence
status	18%
purpose	15%
credit history	8%
property	8%
savings	7%
duration	6%
employment	5%
telephone	5%

Example decisions

Record #2 Rejected

The model rejected this record, mainly because of status (below 0 DM) and purpose (A40), compared with a typical record.

Main factors: status = below 0 DM (hurt), purpose = A40 (hurt), property = A124 (hurt), housing = free (hurt)

What would change the outcome

- If purpose were 'A41' instead of 'A40', the decision would likely be approved.

Record #0 Approved

The model approved this record, mainly because of status (no account) and savings (unknown), compared with a typical record. Property (A124) counted against it, but not enough to change the outcome.

Main factors: status = no account (helped), property = A124 (hurt), savings = unknown (helped), housing = free (hurt)

What would change the outcome

- No single realistic change reverses this decision.

Possible proxy features

Feature	Sensitive attribute	Cramer's V
age	age_band	0.98

8. Limitations

- Fairness metrics can conflict mathematically when groups have different base rates; passing the primary metric does not imply passing every metric.
- Results describe the supplied data only. If outcomes were recorded under a biased historical process, outcome-based metrics inherit that bias.
- Groups with fewer than 100 records are reported with low confidence and are not counted as firm failures.
- Per-decision explanations come from a transparent surrogate model that approximates the audited model; its agreement rate is stated in section 6.
- Counterfactuals change one feature at a time within the observed range and indicate sensitivity; they are not guaranteed routes to a different outcome.
- Proxy detection measures statistical association only and does not show that the model relies on the feature.
- This report supports, but does not replace, legal and domain review.

9. Recommendations

- **Medium** 'age' carries essentially the same information as 'age_band' (Cramer's V 0.98). If the model uses it, the sensitive attribute is effectively a direct input; confirm this is lawful and justified for the use case.
- **Low** 7 group(s) have fewer than 100 people; their results are low-confidence.
- **Low** Schedule a fairness and drift re-check every quarter.

10. Audit trail and sign-off

Report generated	2026-10-07 12:14 UTC
Tool	FairLens 1.2 (Python 3.11.15)
Groups tested	11 (7 with fewer than 100 records)

Role	Name	Signature	Date
Model owner			
Independent reviewer			
Compliance / risk			